Mahmoud Khaled Mohamed

Science Educator & Educational Content Developer

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PROFESSIONAL SUMMARY

Biophysics graduate (Ain Shams University, 2025) with 4+ years of hands-on experience preparing educational material, starting during my university studies. I turn complex science topics into clear, step-by-step lessons: interactive e-books, educational games, slide decks and a full YouTube course. My educational resources on ResearchGate have reached 16,000+ reads. I am seeking a teaching or educational-content role at a school, academy or private university, where I can explain science clearly and build engaging material for students.

EDUCATION

B.S. in Biophysics 10/2021 – 07/2025
Faculty of Science, Ain Shams University
CGPA: 3.435 / 4 (B+)
Graduation project (Excellent): Dopamine Detection Using Carbon Quantum Dots from Banana Peels

EXPERIENCE

Educational Content Developer 2021 – Present
Independent
Prepared educational material for 4+ years, starting during my undergraduate studies: slide decks, study notes, interactive e-books and educational games. Turned textbook chapters into guided, step-by-step lessons with examples, quizzes and answers. Built interactive visuals so students can see ideas (forces, density, crystals, circuits, logic) instead of only reading about them. Published the work openly on ResearchGate, YouTube and itch.io.

SKILLS

Teaching and content
lesson explanation, worked examples, quizzes and answer keys, interactive e-books, educational games, presentations.

Technical
Microsoft Office (Word, PowerPoint, Excel, Power BI), MySQL; basics of Tableau and MATLAB.

Languages
Arabic (native) · English (very good, written and spoken)

EDUCATIONAL MATERIAL AND PROJECTS

Force and Acceleration – Interactive Book
a 7-step guided lesson from "What is a force?" to Newton's second law, with predict-then-check activities, worked examples, a practice generator, exam practice, a quiz and a cheat sheet.

Density – Interactive E-Book (IGCSE Physics)
covers the formula triangle, units, calculations, four ways to measure density, floating and sinking, and a density tower, with check-yourself questions and plain-words explanation prompts.

Crystallography for Beginners – Interactive E-Book
an introduction to crystal structures with interactive 3D visuals for first-time learners.

Galileo Base – Forces & Motion in 3D
32 levels. Students walk through a first-person lab, launch rockets, crash-test cars, balance seesaws and explore orbits. Based on IGCSE forces-and-motion topics.

Volt Valley – Electric Circuits in 3D
48 levels, from Ohm's law and series/parallel circuits to capacitors, coils, AC and filters.

Logic City – Digital Electronics in 3D
32 levels, from binary numbers and logic gates to adders, flip-flops and counters.

Educational Games
free to play on itch.io Every level has a short animated lesson, step-by-step hints and a built-in calculator. I designed the lesson sequence, level goals and explanations for every game.

YouTube course: Crystallography for Beginners
planned, recorded and published a full video course called “Crystallography for Beginners”, breaking the topic into clear, beginner-friendly lessons.

ResearchGate
published educational resources that have reached 16,000+ reads.

COURSES AND WORKSHOPS

McKinsey Forward Program 10/2025 – 12/2025
structured problem-solving, persuasive communication, adaptability.

Business English – Conversation Program 07/2025
ASU Career Center
professional English, presenting and writing.

Up Skill Program 07/2024
ASU Career Center
presentation, communication and teamwork skills.